How Crossing Over Creates New Allele Combinations

Biology 2e chapter 11 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

Homologous chromosomes carry the same genes at corresponding positions and can carry different alleles. After DNA replication, each homolog has two sister chromatids. During prophase I, corresponding DNA segments can be exchanged between nonsister chromatids.

Investigation question

How does exchanging chromosome segments change the allele combinations in gametes?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

Build a paired set of replicated ABC and abc chromosomes. Select one chromatid from each homolog and the assigned exchange position. Follow the chromosomes through both divisions, place all four chromatids in gametes, then tally parental and recombinant allele combinations. Compare three models before exploring.

Align the homologs. For each exchange model select one nonsister chromatid from each homolog, then the assigned cut position. Exchange matching end segments. Separate homologs, then place each sister chromatid in a gamete from its own cell. Record all four allele strings and tally parental and recombinant products. Review observations, then click Collect data and continue. Complete the assigned comparison before exploring.

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Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: gametes

Condition / sample	Repeat	Source	Parental gametes	Recombinant gametes
No exchange comparison	1			
Exchange between A and B	1			
Exchange between B and C	1			

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Sample / field: _____ Source: _____

Item / band / cell	Measurement or classification	Reasoning / uncertainty

For large counts, use tallies with a clear starting position so each specimen is counted once. For microscopy and gels, record calibration and measurement units.

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Compare the four gametes from each model. Which allele combinations are new relative to ABC and abc?

Why must corresponding segments be exchanged between nonsister chromatids? Explain why this changes combinations without creating new alleles.

How did moving the crossover point change the gametes? Explain one limitation of this simplified model.

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: A manipulable chromosome model, not observed cell data. One homologous pair, three loci, and at most one crossover are represented. The no-exchange comparison isolates the effect of recombination; real meiosis includes additional chromosome pairs, assortment and crossover constraints. The model does not predict real recombination frequencies.

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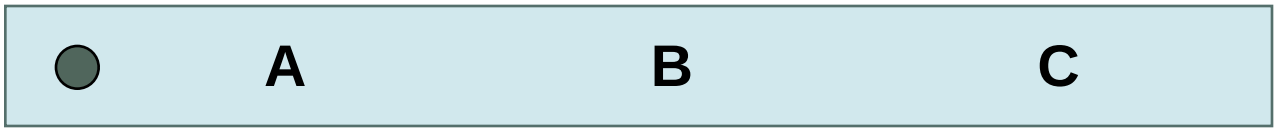
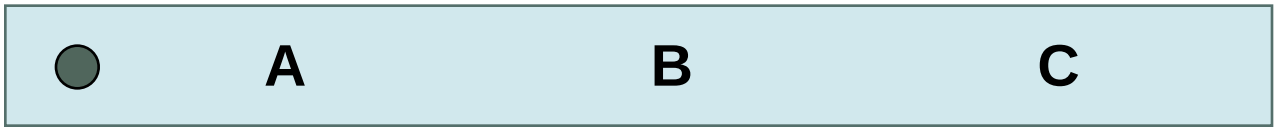
Biology 2e chapter 11 reading connection | Printable chromosome strips

Print one sheet for each model, or use removable tape to reuse strips. Cut around each strip. The dot is the centromere. Matching letters mark corresponding gene positions. Pair the two same-color strips as sister chromatids.

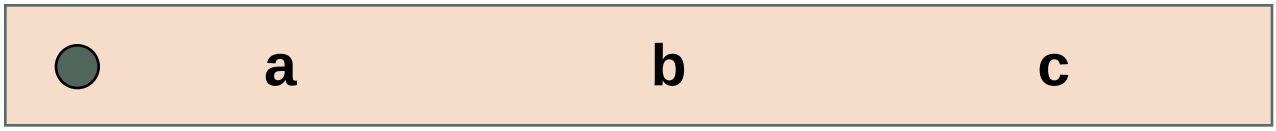
For crossing over, cut ONE strip of each color at the same gap between genes. Exchange the complete right-hand segments. Keep the other two strips unchanged.



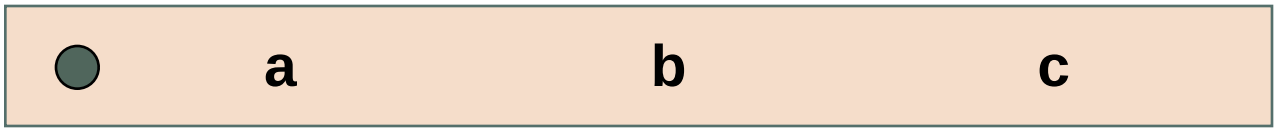
Homolog 1 / chromatid 1



Homolog 1 / chromatid 2



Homolog 2 / chromatid 3



Homolog 2 / chromatid 4

Draw four gamete circles on another sheet. Separate homologs first, then sisters. Place one resulting strip in each circle; record its allele string and whether it is parental or recombinant.

Optional evidence extension: the separate Sordaria activity connects chromosome modeling to ordered spore patterns.